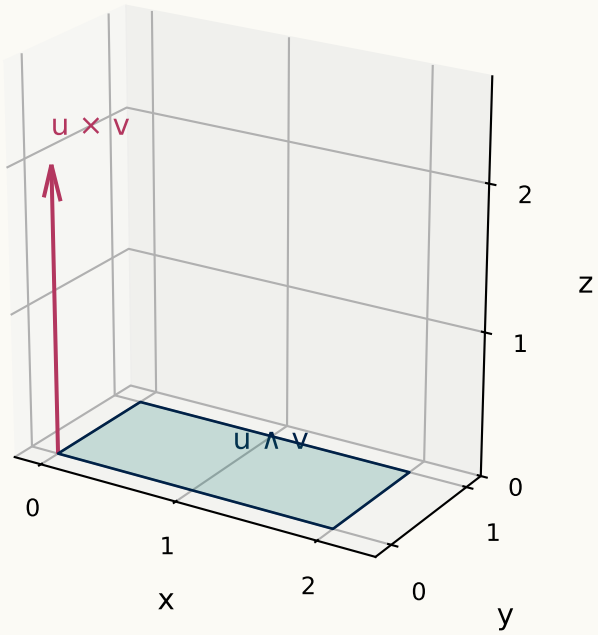In $\mathbb{R}^3$ a normal arrow can encode an oriented patch